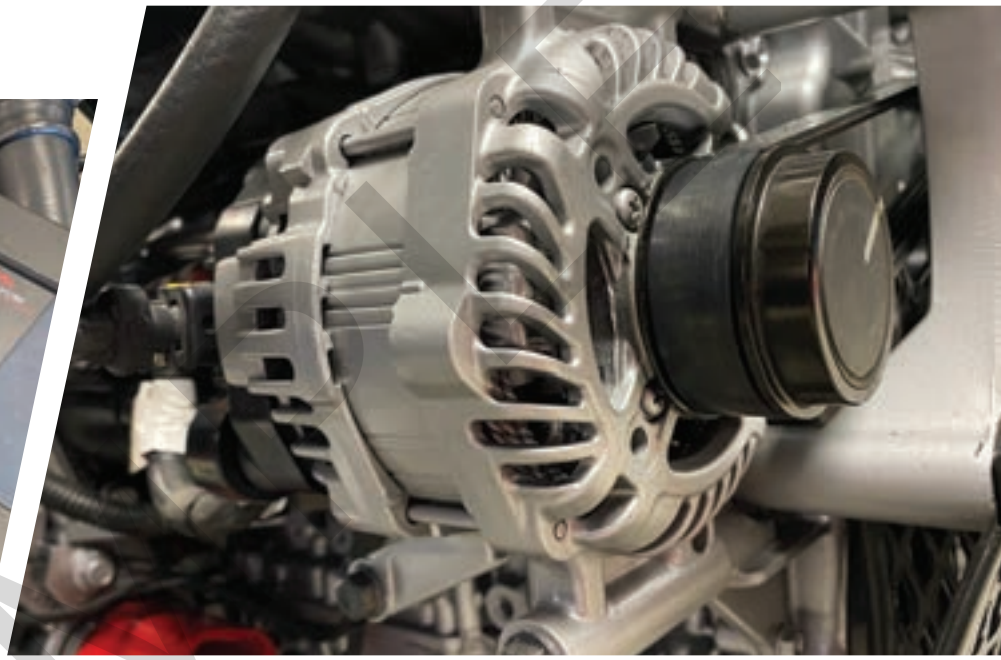
Auto Electricity and Electronics

EIGHTH EDITION



James E. Duffy | Chris McNally

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Preface

Automobiles and light trucks are now as much electronic as they are mechanical. Look under the hood of any late-model car and you will find a maze of wires connecting dozens of devices. In fact, electronics can be found in almost every major system of a car, including the suspension, transmission, fuel, ignition, emission control, and braking systems. Some cars have more than one hundred separate on-board computers and electronic control modules.

The day of the "backyard mechanic" is over; this is the era of the well-trained electronic technician. Conventional electrical systems, when combined with newer computer-controlled systems, make it necessary to have a sound background in electricity and electronics. This knowledge will allow you to do competent repair work. Anyone versed in auto electronics will have no problem finding and keeping a good job in the automotive service industry.

Auto Electricity and Electronics was designed to make the complex seem simple. Short sentences, abundant illustrations and photos, and easy-to-follow procedures are used to make learning easier and more fun. All technical terms are printed in ***bold italic*** and clearly defined so that you can quickly learn the language of the automotive electronic technician.

The text is divided into 35 chapters. The first chapters review fundamentals of electricity and electronics: electrical principles and components, electronic components, wiring repairs, soldering, wiring diagrams, special tools and equipment, basic electrical tests, and safety. The main body of the book provides information about the construction and operation of major electrical and electronic systems, and how to diagnose and repair issues in those systems. The last section covers ASE and career information, useful tables, a glossary, and an index.

New chapters on battery electric vehicles bring this text up to date in describing the latest innovations in auto electricity and electronics. Extensive updates throughout reflect the increasing importance of electricity and electronics to modern vehicles in almost every system, from starting to entertainment and communications. *Auto Electricity and Electronics* stresses the use of on-board diagnostics, as well as conventional meter tests, to find electrical and electronic problems in all major automotive systems. Information on OBD II diagnostic systems has been strategically placed throughout the text. All electrical and electronic terminology has been updated to conform to the latest SAE standards (J1930).

A shop manual available for use with the *Auto Electricity and Electronics* textbook contains hands-on jobs that allow you to practice the service and repair procedures detailed in the text. The *Auto Electricity and Electronics* text and the accompanying shop manual are available in both print and digital formats, and are supported with professionally developed, user-friendly online resources. These resources include study tools, videos, and activities designed to prepare you for success in your studies and career. We encourage you to take advantage of all the resources available as part of the *Auto Electricity and Electronics* learning solution.

Auto Electricity and Electronics is a valuable tool for anyone wanting to know more about modern automobiles. It will help the experienced technician learn about new technology, review principles, and prepare for ASE certification. It will guide the student through a sequential program of study to prepare for employment. It will also help the car owner who just wants to know more about what is going on inside all those "black boxes."

James Duffy
Chris McNally

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About the Authors

James E. Duffy has devoted much of his professional life to developing educational materials that help people understand vocational and technical subjects. Mr. Duffy earned a Bachelor of Science degree in Industrial Technology and an Associate of Science degree in Industrial Electronics from Indiana State University. After college, Mr. Duffy worked as an automotive instructor at Brandon High School in Brandon, Florida, and at Hillsborough Adult Education Center in Tampa, Florida. After teaching for several years, Mr. Duffy joined Goodheart-Willcox Publisher as a technical editor and eventually began writing textbooks. In addition to *Auto Electricity and Electronics*, Mr. Duffy's books include *Auto Engine Repair*, *Automatic Transmissions and Transaxles*, *Manual Drive Trains and Axles*, and *Modern Automotive Technology*.

Dr. Chris McNally is the chair of the Applied Technologies Department at Hudson Valley Community College. In this role, he oversees students in degree and certificate programs in Automotive Management, Automotive Technical Services, Automotive Technical Services–Autobody Repair, Clean Energy Management, Electric and Autonomous Vehicles, Electrical Construction and Maintenance, Heating/Air Conditioning/Refrigeration Technical Services, Overhead Electric Line Worker, and Mechatronics.

Dr. McNally, who received the State University of New York Chancellor's Award for Excellence in Teaching in 2009, has been a faculty member teaching automotive technology at the college since 1996 and recently led efforts to create the new associate degree program in Autonomous and Electric Vehicles. He also oversees a National Science Foundation (NSF) Advanced Technology Education grant related to educating automotive and electrical construction students about electric vehicle charging systems.

A two-time finisher of the Lake Placid Ironman Triathlon, he is the founder of IOU (It's on Us), a nonprofit, grassroots movement to raise awareness of the problem of sexual harassment of female athletes.

Dr. McNally received his doctoral and master's degrees from the State University of New York at Albany, his baccalaureate degree from SUNY Oswego, and his associate degree in automotive technology from Herkimer County Community College.

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The authors and publisher wish to thank the following industry and teaching professionals for their valuable input into the development of *Auto Electricity and Electronics*.

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TIF Instruments
Toyota
TRW
Union Electronics
Universal Sound
Volvo

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New to This Edition

The eighth edition of *Auto Electricity and Electronics* contains numerous enhancements to help you succeed. It has been thoroughly revised by the authors and carefully vetted by a panel of experts in the field of automotive education to make sure the material is technically accurate, up to date, and presented in the most logical and effective manner. Content has been updated throughout the text to feature the latest technology, techniques, tools, and equipment. Significant updates include the following:

- Content has been updated to ensure alignment with all three levels of the latest ASE Education Foundation Engine Repair Task List (MLR, AST, and MAST).
- Content has been edited with an eye toward relevancy for a focused, up-to-date product designed to provide you with the knowledge and skills needed to succeed as an automotive technician.
- New chapters have been added to focus on battery electric vehicles.
- The chapter sequence has been revised so the diagnosis and service chapters immediately follow the corresponding theory chapters where applicable.
- End-of-chapter questions have been enhanced to include Know and Understand, Apply and Analyze, and Critical Thinking questions, as well as ASE-Type questions.
- New images have been incorporated throughout the text to show the latest tools, equipment, and diagnostic techniques.
- Requested by automotive students, a new Shop Talk feature located at the beginning of each chapter explains why the material covered in the chapter is relevant to the study of auto electricity and electronics.
- The exciting new interior design makes it easy to navigate content, with important terms and figure references highlighted within the text.

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ASE Connections

We Support

Education Foundation

Auto Electricity and Electronics is correlated to standards established by the National Institute for Automotive Service Excellence (ASE) and the ASE Education Foundation. These standards were developed with input from industry experts to ensure they reflect the skills students and technicians must master to succeed in the automobile service and repair industry.

ASE certification is the benchmark for automobile repair technicians. It is recommended that you seek ASE certification once you have sufficient knowledge and meet the work experience requirements. Visit the ASE website for more information on ASE certification.

The ASE Education Foundation has established a training program accreditation process designed to improve the quality of training offered at both the secondary and postsecondary levels. The ASE Education Foundation grants accreditation to programs that comply with their evaluation procedures, meet established standards, and adhere to the policies set forth by the organization. To learn more about the accreditation process, visit the ASE Education Foundation website.

To see how *Auto Electricity and Electronics* correlates to credentialing and certification standards, visit the Correlations tab at www.g-w.com/auto-electricity-electronics-2026.

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FEATURES OF THE TEXTBOOK

The instructional design of this textbook includes student-focused learning tools to help you succeed. This visual guide highlights these features.

Chapter Opening Materials

Each chapter opener contains a list of Learning Outcomes, a list of Technical Terms, a Shop Talk feature, and an Introduction.

Learning Outcomes clearly identify the knowledge and skills to be gained when the chapter is completed.

Technical Terms list the key words to be learned in the chapter.

The **Shop Talk** feature explains why the material covered in the chapter is relevant to the study of automotive electricity and electronics and how it is relevant to the diagnosis and service of automotive electrical systems.

The **Introduction** provides an overview and preview of the chapter content.

Illustrations

Illustrations have been designed to clearly and simply communicate the specific topic. Illustrations and photographic images have been updated throughout to show the latest vehicles, equipment, and techniques.

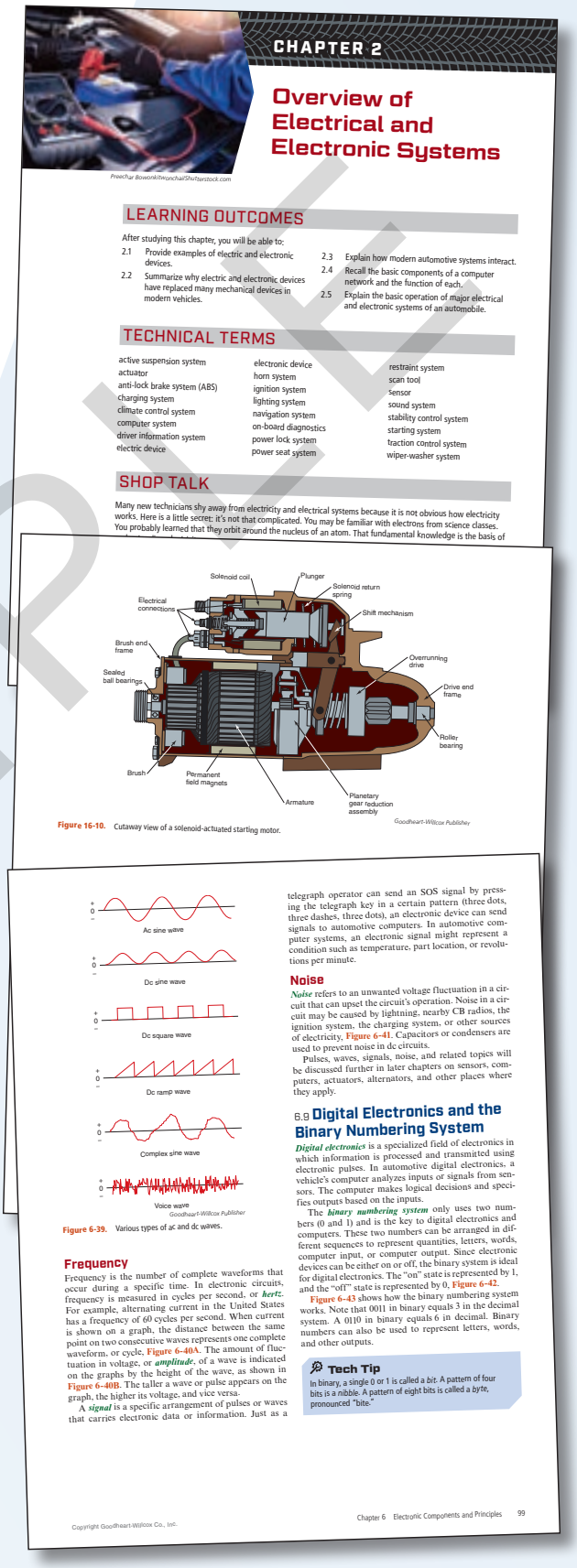
Additional Features

Additional features are used throughout the body of each chapter to further learning and knowledge.

Tech Tips provide supplemental technical information and service hints related to the procedure or system being explained.

Warnings alert you to potentially dangerous materials and practices that can result in personal injury if proper procedures or safety measures are not followed.

Cautions identify situations that may cause damage to a vehicle, equipment, or tools if proper procedures are not followed.



CHAPTER 2

Overview of Electrical and Electronic Systems

LEARNING OUTCOMES

After studying this chapter, you will be able to:

2.1 Provide examples of electric and electronic devices.
2.2 Summarize why electric and electronic devices have replaced many mechanical devices in modern vehicles.
2.3 Explain how modern automotive systems interact.
2.4 Recall the basic components of a computer network and the function of each.
2.5 Explain the basic operation of major electrical and electronic systems of an automobile.

TECHNICAL TERMS

active suspension system, actuator, anti-lock brake system (ABS), charging system, climate control system, computer system, driver information system, electric device, electronic device, horn system, ignition system, lighting system, navigation system, on-board diagnostics, power lock system, power seat system, restraint system, scan tool, sensor, sound system, stability control system, starting system, traction control system, wiper-washer system

SHOP TALK

Many new technicians shy away from electricity and electrical systems because it is not obvious how electricity works. Here is a little secret: it's not that complicated. You may be familiar with electrons from science classes. You probably learned that they orbit around the nucleus of an atom. That fundamental knowledge is the basis of

Figure 16-10. Cutaway view of a solenoid-actuated starting motor.

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Figure 6-39. Various types of ac and dc waves.

Frequency

Noise

6.9 Digital Electronics and the Binary Numbering System

Tech Tip

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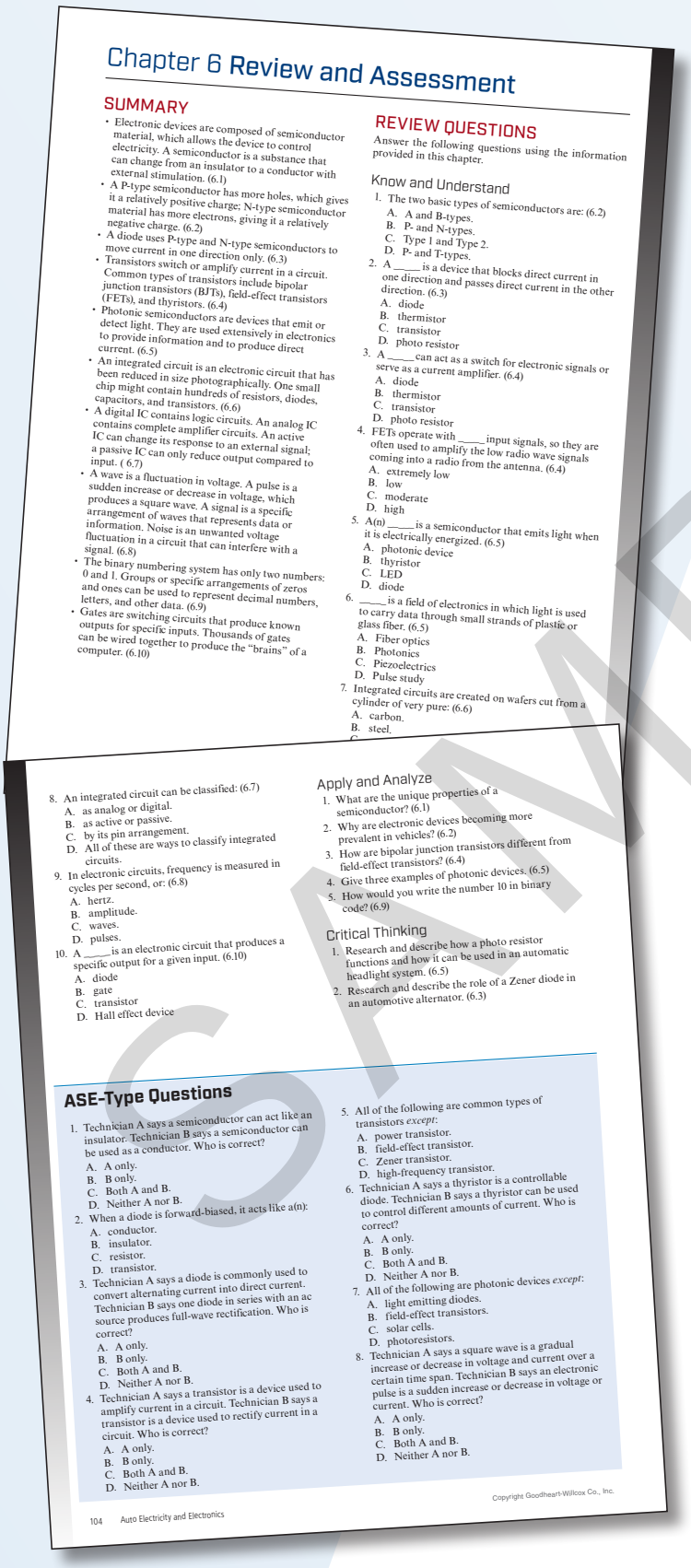

Chapter 6 Review and Assessment

SUMMARY

- Electronic devices are composed of semiconductor material, which allows the device to control electricity. A semiconductor is a substance that can change from an insulator to a conductor with external stimulation. (6.1)
- A P-type semiconductor has more holes, which gives it a relatively positive charge; N-type semiconductor material has more electrons, giving it a relatively negative charge. (6.2)
- A diode uses P-type and N-type semiconductors to move current in one direction only. (6.3)
- Transistors switch or amplify current in a circuit. Common types of transistors include bipolar junction transistors (BJTs), field-effect transistors (FETs), and thyristors. (6.4)
- Photonic semiconductors are devices that emit or detect light. They are used extensively in electronics to provide information and to produce direct current. (6.5)
- An integrated circuit is an electronic circuit that has been reduced in size photographically. One small chip might contain hundreds of resistors, diodes, capacitors, and transistors. (6.6)
- A digital IC contains logic circuits. An analog IC contains complete amplifier circuits. An active IC can change its response to an external signal; a passive IC can only reduce output compared to input. (6.7)
- A wave is a fluctuation in voltage. A pulse is a sudden increase or decrease in voltage, which produces a square wave. A signal is a specific arrangement of waves that represents data or information. Noise is an unwanted voltage fluctuation in a circuit that can interfere with a signal. (6.8)
- The binary numbering system has only two numbers: 0 and 1. Groups or specific arrangements of zeros and ones can be used to represent decimal numbers, letters, and other data. (6.9)
- Gates are switching circuits that produce known outputs for specific inputs. Thousands of gates can be wired together to produce the "brains" of a computer. (6.10)

REVIEW QUESTIONS

Answer the following questions using the information provided in this chapter.

Know and Understand

1. The two basic types of semiconductors are: (6.2)
   A. A and B-types.
   B. P- and N-types.
   C. Type 1 and Type 2.
   D. P- and T-types.
2. A _____ is a device that blocks direct current in one direction and passes direct current in the other direction. (6.3)
   A. diode
   B. thermistor
   C. transistor
   D. photo resistor
3. A _____ can act as a switch for electronic signals or serve as a current amplifier. (6.4)
   A. diode
   B. thermistor
   C. transistor
   D. photo resistor
4. FETs operate with _____ input signals, so they are often used to amplify the low radio wave signals coming into a radio from the antenna. (6.4)
   A. extremely low
   B. low
   C. moderate
   D. high
5. A(n) _____ is a semiconductor that emits light when it is electrically energized. (6.5)
   A. photonic device
   B. thyristor
   C. LED
   D. diode
6. _____ is a field of electronics in which light is used to carry data through small strands of plastic or glass fiber. (6.5)
   A. Fiber optics
   B. Photonics
   C. Piezoelectrics
   D. Pulse study
7. Integrated circuits are created on wafers cut from a cylinder of very pure: (6.6)
   A. carbon.
   B. steel.
   C.
8. An integrated circuit can be classified: (6.7)
   A. as analog or digital.
   B. as active or passive.
   C. by its pin arrangement.
   D. All of these are ways to classify integrated circuits.
9. In electronic circuits, frequency is measured in cycles per second, or: (6.8)
   A. hertz.
   B. amplitude.
   C. waves.
   D. pulses.
10. A _____ is an electronic circuit that produces a specific output for a given input. (6.10)
   A. diode
   B. gate
   C. transistor
   D. Hall effect device

Apply and Analyze

1. What are the unique properties of a semiconductor? (6.1)
2. Why are electronic devices becoming more prevalent in vehicles? (6.2)
3. How are bipolar junction transistors different from field-effect transistors? (6.4)
4. Give three examples of photonic devices. (6.5)
5. How would you write the number 10 in binary code? (6.9)

Critical Thinking

1. Research and describe how a photo resistor functions and how it can be used in an automatic headlight system. (6.5)
2. Research and describe the role of a Zener diode in an automotive alternator. (6.3)

ASE-Type Questions

1. Technician A says a semiconductor can act like an insulator. Technician B says a semiconductor can be used as a conductor. Who is correct?
   A. A only.
   B. B only.
   C. Both A and B.
   D. Neither A nor B.
2. When a diode is forward-biased, it acts like a(n):
   A. conductor.
   B. insulator.
   C. resistor.
   D. transistor.
3. Technician A says a diode is commonly used to convert alternating current into direct current. Technician B says one diode in series with an ac source produces full-wave rectification. Who is correct?
   A. A only.
   B. B only.
   C. Both A and B.
   D. Neither A nor B.
4. Technician A says a transistor is a device used to amplify current in a circuit. Technician B says a transistor is a device used to rectify current in a circuit. Who is correct?
   A. A only.
   B. B only.
   C. Both A and B.
   D. Neither A nor B.
5. All of the following are common types of transistors *except*:
   A. power transistor.
   B. field-effect transistor.
   C. Zener transistor.
   D. high-frequency transistor.
6. Technician A says a thyristor is a controllable diode. Technician B says a thyristor can be used to control different amounts of current. Who is correct?
   A. A only.
   B. B only.
   C. Both A and B.
   D. Neither A nor B.
7. All of the following are photonic devices *except*:
   A. light emitting diodes.
   B. field-effect transistors.
   C. solar cells.
   D. photoresistors.
8. Technician A says a square wave is a gradual increase or decrease in voltage and current over a certain time span. Technician B says an electronic pulse is a sudden increase or decrease in voltage or current. Who is correct?
   A. A only.
   B. B only.
   C. Both A and B.
   D. Neither A nor B.

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End-of-Chapter Content

End-of-chapter material provides an opportunity for review and application of concepts.

A concise **Summary** provides an additional review tool and reinforces key learning objectives. This helps you focus on important concepts presented in the text.

Know and Understand questions enable you to demonstrate knowledge, identification, and comprehension of chapter material.

Apply and Analyze questions extend learning and develop your abilities to use learned material in new situations and to break down material into its component parts.

Critical Thinking questions develop higher-order thinking, problem-solving, personal, and workplace skills.

ASE-Type Questions are designed to help you become familiar with the types of questions you will encounter when taking the ASE certification tests.

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TOOLS FOR STUDENT AND INSTRUCTOR SUCCESS

Student Tools

Auto Electricity and Electronics teaches students the operation, diagnosis, and service of electrical and electronic systems that are found in modern automobiles, including battery systems, charging systems, lighting and instrumentation systems, and hybrid vehicle systems. This textbook emphasizes the use of strategy-based troubleshooting procedures and the latest diagnostic equipment to pinpoint problems. It is a valuable resource for anyone who needs a thorough understanding of automotive electricity and electronics, including those preparing for ASE certification test A6, Electrical/Electronic Systems.

Shop Manual

- Provides hands-on practice of 100% of the auto electricity and electronics tasks in all three levels of the ASE Education Foundation Task List (MLR, AST, and MAST).
- Each job in the shop manual is a hands-on activity that corresponds to one or more ASE Education Foundation Tasks.
- The jobs help students develop the hands-on skills needed to succeed in the field and are intended to be completed in a shop setting under the guidance of an instructor.

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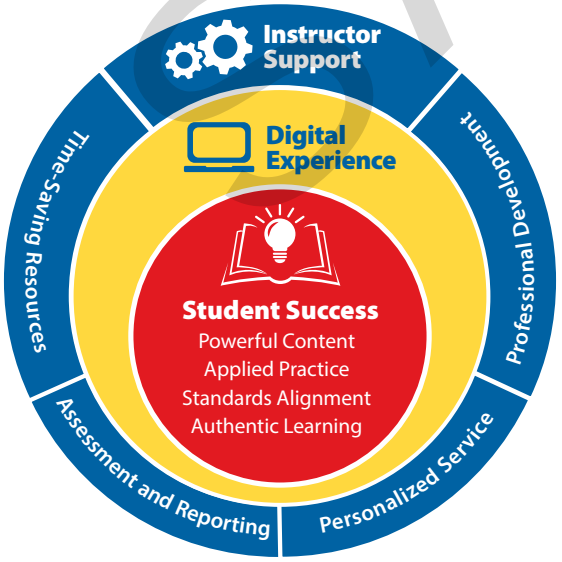

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- The **Image Library** includes images from throughout the text for review, reference, or for customizing and creating additional supplements.

See www.g-w.com/auto-electricity-electronics-2026 for a list of all available formats.

Professional Development

- Expert content specialists
- Research-based pedagogy and instructional practices
- Options for virtual and in-person professional development

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